

Primary VPT Chain: ProducerUA -> ProZorro -> Retail

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely I(1)-like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

standardized_type=sour_cream

retailer=silpo

promo_variant=promo_controlled

frequency=daily

Primary chain is strictly ProducerUA -> ProZorro -> Retail. Retail is estimated separately for silpo, novus, and silpo_novus. Combined rule: daily median of available silpo and novus standardized_type prices.

Table: PreTests

series	adf_level_p	kpss_level_p	adf_diff1_p	kpss_diff1_p	adf_diff2_p	kpss_diff2_p	integration_class	stability_flag	standardized_type	retailer	promo_variant	cointegration_p	pair_n_obs
producer	0.01408153516990551	0.01	4.706435460152205e-13	0.1	3.085032729165153e-21	0.1	I(1)	1.0	sour_cream	silpo	promo_controlled	nan	nan
prozorro	0.21880956753864866	0.04947047925942609	2.411890357322739e-12	0.1	3.969186107841103e-13	0.1	I(1)	0.0	sour_cream	silpo	promo_controlled	nan	nan
retail	0.0001866404404338293	0.1	1.2992938855568006e-06	0.1	1.670971302457964e-10	0.1	I(0)	0.0	sour_cream	silpo	promo_controlled	nan	nan
producer_to_prozorro	nan	nan	nan	nan	nan	nan	pair	nan	sour_cream	silpo	promo_controlled	0.3418935830778413	178.0
prozorro_to_retail	nan	nan	nan	nan	nan	nan	pair	nan	sour_cream	silpo	promo_controlled	0.0009999496932936124	38.0
producer_to_retail	nan	nan	nan	nan	nan	nan	pair	nan	sour_cream	silpo	promo_controlled	0.9914737446395873	48.0

Table: ModelCoefficients

standardized_type	retailer	promo_variant	frequency	link	model_family	y_series	x_series	n_obs	sr_coef	lr_coef	ect_coef	ect_pvalue	asymmetry_short_p	asymmetry_long_p	vecm_rank	model_status	unreliable_flag	notes	primary_chain_doc
sour_cream	silpo	promo_controlled	daily	producer_to_prozorro	ARDL	prozorro	producer	178	0.15033213274852208	0.532018058976545	nan	nan	nan	nan	nan	ok	0	bounds_lower_p=0.1226; bo...	Primary chain is strictly...
sour_cream	silpo	promo_controlled	daily	producer_to_prozorro	NARDL	prozorro	producer	176	0.8049124174362638	0.31818036940691613	-0.9103202976511392	1.293616851986384e-25	0.2798213093974461	0.3401073440796145	nan	unreliable	1	Wald short/long asymmetry...	Primary chain is strictly...
sour_cream	silpo	promo_controlled	daily	prozorro_to_retail	ARDL	retail	prozorro	38	-0.09602698663316289	-0.07808211302448108	nan	nan	nan	nan	nan	ok	0	bounds_lower_p=nan; bound...	Primary chain is strictly...
sour_cream	silpo	promo_controlled	daily	prozorro_to_retail	NARDL	retail	prozorro	36	-0.025916997864596635	-0.022225374920711808	-0.8330689213690317	0.00037765858297258574	0.7464656966292003	0.8267608548502503	nan	ok	0	Wald short/long asymmetry...	Primary chain is strictly...

Table: ResidualDiagnostics

model_family	link	y_series	x_series	ljungbox_p	arch_p	jb_p	unreliable_flag
ARDL	producer_to_prozorro	prozorro	producer	0.9991919902903723	0.4332665662563698	0.028152037118156784	0
NARDL	producer_to_prozorro	prozorro	producer	0.07252573667137008	0.012783066753899843	0.032619696848474544	1
ARDL	prozorro_to_retail	retail	prozorro	0.9055231898701895	1.0	3.661771980151874e-23	0
NARDL	prozorro_to_retail	retail	prozorro	0.7865388890305405	0.33444920538763856	0.29874884459634815	0

Table: SeriesUsed

date	producer	prozorro	retail	standardized_type	retailer	promo_variant	combined_rule
2021-02-01	12.7168318078645	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-02	12.72632554286918	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-03	12.73569490777093	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-04	12.74493705667296	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-05	12.75404914367847	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-06	12.76302832289066	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-07	12.77187174841275	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-08	12.78057657434793	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-09	12.78861583935331	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-10	12.79582084628455	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-11	12.80272614554264	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-12	12.80986628752861	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-13	12.81777582264346	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-14	12.8269893012882	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-15	12.83804127386385	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-16	12.85602638141248	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-17	12.88286036370906	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-18	12.91430941925396	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-19	12.94613974654752	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-20	12.97411754409009	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-21	12.99400901038205	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-22	13.00158034392373	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-23	12.85209968138777	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-24	12.46659692011084	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-25	11.93948089958932	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-26	11.3651604593196	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-27	10.83804443879808	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-02-28	10.45254167752114	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-03-01	10.30306101498519	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-03-02	10.3211976511164	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-03-03	10.367971081139	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-03-04	10.43192658749644	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-03-05	10.50160945263215	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-03-06	10.56556495898959	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-03-07	10.61233838901219	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-03-08	10.6304750251434	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-03-09	10.62692996199335	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-03-10	10.6172401963284	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-03-11	10.60282386382634	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...
2021-03-12	10.58509910016495	nan	nan	sour_cream	silpo	promo_controlled	daily_median_of_available...

Table: LagProfile

	pair	lag	corr
producer_to_prozorro	1	0.07433869937437788	
producer_to_prozorro	2	0.06992193974139833	
producer_to_prozorro	3	0.05247780353970976	
producer_to_prozorro	4	0.03707093523380621	
producer_to_prozorro	5	0.04488476174629457	
producer_to_prozorro	6	0.03942159255359992	
producer_to_prozorro	7	0.04774350069192749	
producer_to_prozorro	8	0.055533922878419725	
producer_to_prozorro	9	0.05815494362517447	
producer_to_prozorro	10	0.05835686030753	
producer_to_prozorro	11	0.05572808262664563	
producer_to_prozorro	12	0.05132684090763728	
producer_to_prozorro	13	0.0478799443885565	
producer_to_prozorro	14	0.0459078796867533	
producer_to_prozorro	15	0.04627227965570159	
producer_to_prozorro	16	0.04980743752852967	
producer_to_prozorro	17	0.05457381656925348	
producer_to_prozorro	18	0.05403494026121064	
producer_to_prozorro	19	0.045342075346488105	
producer_to_prozorro	20	0.030030963601459588	
producer_to_prozorro	21	0.011140301716297355	
producer_to_prozorro	22	-0.006115614807159968	
producer_to_prozorro	23	-0.017626157690186797	
producer_to_prozorro	24	-0.022831222527256615	
producer_to_prozorro	25	-0.024681880076761423	
producer_to_prozorro	26	-0.028172449275670572	
producer_to_prozorro	27	-0.03484368926283701	
producer_to_prozorro	28	-0.04221286356527852	
producer_to_prozorro	29	-0.04827032698871696	
producer_to_prozorro	30	-0.05291303035138733	
prozorro_to_retail	1	-0.03667757521279544	
prozorro_to_retail	2	-0.27789900612579105	
prozorro_to_retail	3	-0.22461696407750467	
prozorro_to_retail	4	-0.18260700540033242	
prozorro_to_retail	5	-0.019443231840724415	
prozorro_to_retail	6	-0.09530230617623504	
prozorro_to_retail	7	0.11332668653099863	
prozorro_to_retail	8	-0.13332162162664174	
prozorro_to_retail	9	-0.024746620429041553	
prozorro_to_retail	10	0.09934959399577038	

Table: ModelEligibility

standardized_type	retailer	promo_variant	frequency	integration_producer	integration_prozorro	integration_retail	n_obs_producer	n_obs_prozorro	n_obs_retail	n_obs_pair_prod_prozorro	n_obs_pair_prozorro_retail	any_i2	eligibility_note
sour_cream	silpo	promo_controlled	daily	I(1)	I(1)	I(0)	1793	188	48	178	38	0	Admissible for ARDL/ECM s...

Table: NARDL_Multipliers

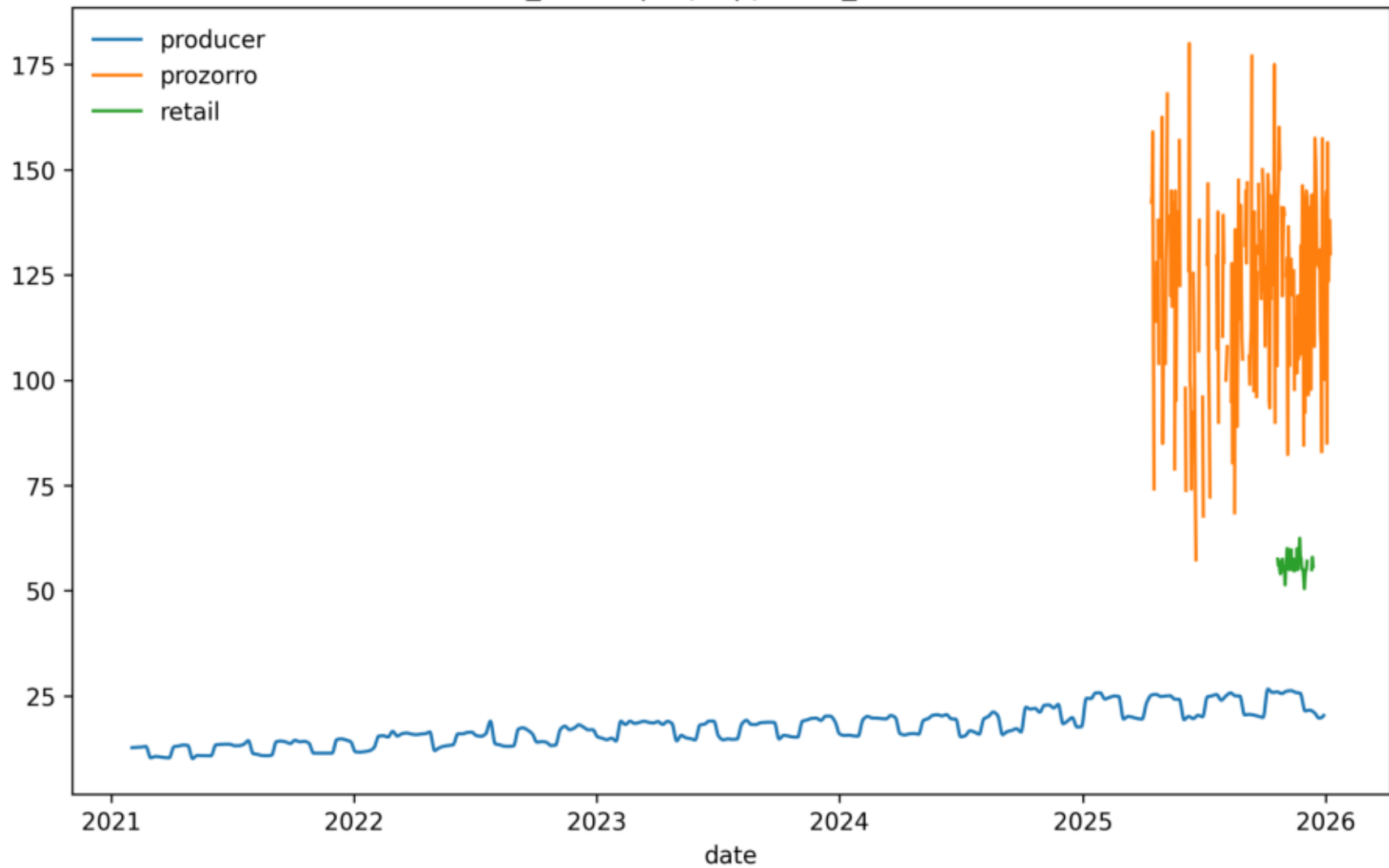
horizon	mult_pos	mult_neg	link
0	0.5272629908128266	-0.2776494266234373	producer_to_prozorro
1	0.5660151486488386	-0.2980557790340301	producer_to_prozorro
2	0.5688633093639158	-0.29955558122620274	producer_to_prozorro
3	0.5690726401447522	-0.29966581193134717	producer_to_prozorro
4	0.5690880252933408	-0.2996739135386258	producer_to_prozorro
5	0.5690891560529747	-0.29967450898107	producer_to_prozorro
6	0.569089239160222	-0.29967455274420146	producer_to_prozorro
7	0.5690892452683411	-0.29967455596065284	producer_to_prozorro
8	0.5690892457172686	-0.29967455619705186	producer_to_prozorro
9	0.5690892457502633	-0.2996745562144264	producer_to_prozorro
10	0.5690892457526884	-0.2996745562157034	producer_to_prozorro
11	0.5690892457528666	-0.29967455621579725	producer_to_prozorro
12	0.5690892457528797	-0.29967455621580413	producer_to_prozorro
13	0.5690892457528807	-0.29967455621580463	producer_to_prozorro
14	0.5690892457528807	-0.2996745562158047	producer_to_prozorro
15	0.5690892457528807	-0.2996745562158047	producer_to_prozorro
16	0.5690892457528807	-0.2996745562158047	producer_to_prozorro
17	0.5690892457528807	-0.2996745562158047	producer_to_prozorro
18	0.5690892457528807	-0.2996745562158047	producer_to_prozorro
19	0.5690892457528807	-0.2996745562158047	producer_to_prozorro
20	0.5690892457528807	-0.2996745562158047	producer_to_prozorro
0	0.003140200025901027	0.02905719789049766	prozorro_to_retail
1	0.0030683005116296557	0.028391890458747533	prozorro_to_retail
2	0.0030699467571267004	0.0284071236543368	prozorro_to_retail
3	0.0030699090639063216	0.028406774867840833	prozorro_to_retail
4	0.0030699099269482343	0.02840678285382244	prozorro_to_retail
5	0.0030699099071876163	0.028406782670971603	prozorro_to_retail
6	0.0030699099076400647	0.02840678267515824	prozorro_to_retail
7	0.0030699099076297053	0.028406782675062383	prozorro_to_retail
8	0.003069909907629942	0.028406782675064576	prozorro_to_retail
9	0.003069909907629937	0.028406782675064527	prozorro_to_retail
10	0.003069909907629937	0.028406782675064527	prozorro_to_retail
11	0.003069909907629937	0.028406782675064527	prozorro_to_retail
12	0.003069909907629937	0.028406782675064527	prozorro_to_retail
13	0.003069909907629937	0.028406782675064527	prozorro_to_retail
14	0.003069909907629937	0.028406782675064527	prozorro_to_retail
15	0.003069909907629937	0.028406782675064527	prozorro_to_retail
16	0.003069909907629937	0.028406782675064527	prozorro_to_retail
17	0.003069909907629937	0.028406782675064527	prozorro_to_retail
18	0.003069909907629937	0.028406782675064527	prozorro_to_retail

Table: VECM_IRF

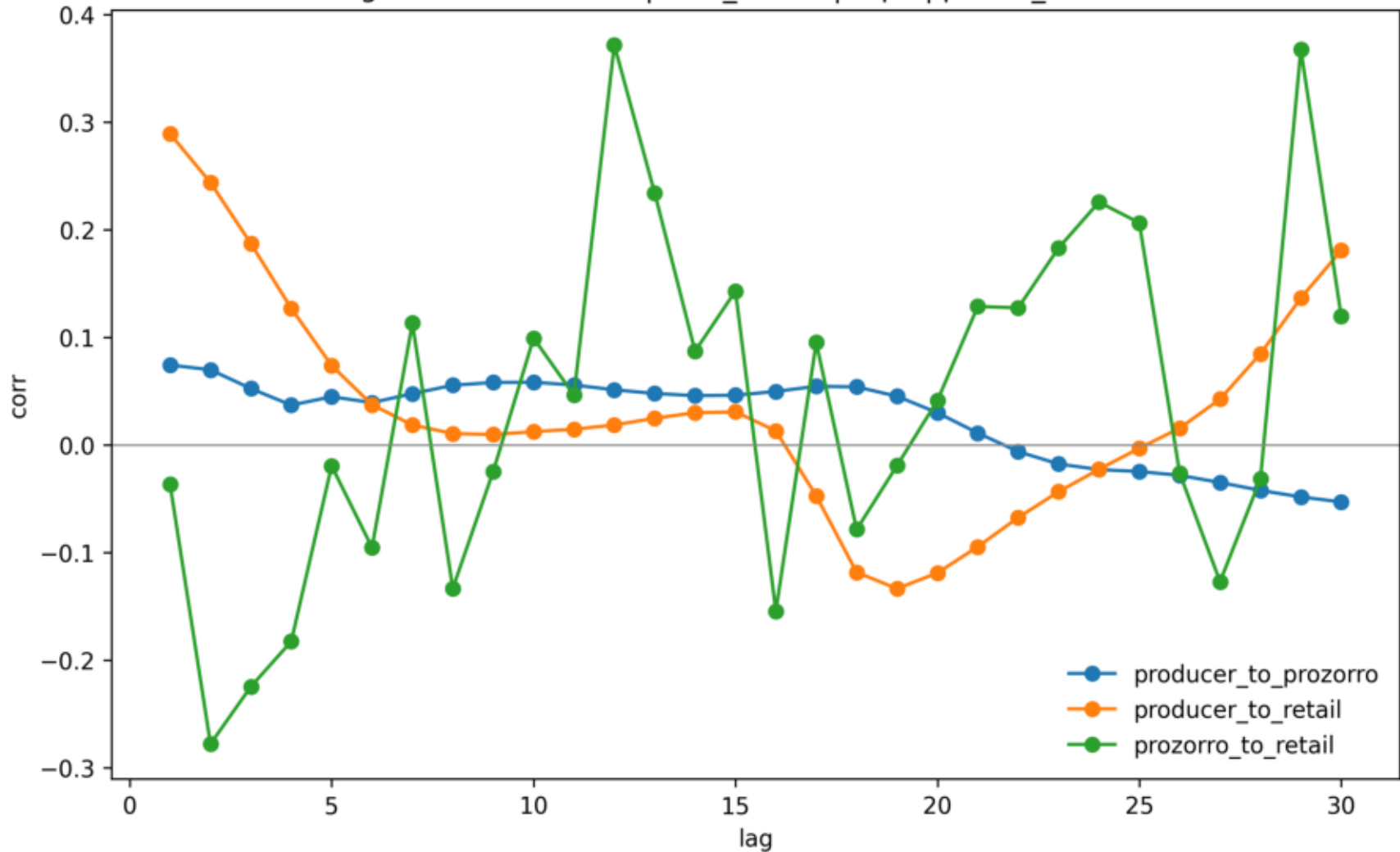
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time_series_promo_controlled.png

sour_cream | silpo | promo_controlled



Lag-Correlation Profile | sour_cream | silpo | promo_controlled



NARDL Dynamic Multipliers | sour_cream | silpo | promo_controlled

